

Programming Assignment 02: OOP

Data Structure and Database
Shiraz University

January 2, 2026

1 Cookie Jar

Suppose that you'd like to implement a cookie jar in which to store cookies. In a file called `jar.py`, implement a class called `Jar` with the following methods:

`__init__` should initialize a cookie jar with the given `capacity`, which represents the maximum number of cookies that can fit in the cookie jar. If `capacity` is not a non-negative integer, `__init__` should raise a `ValueError`.

`__str__` should return a string with n (cookie) symbols, where n is the number of cookies in the cookie jar. For instance, if there are 3 cookies in the cookie jar, then `str` should return `"(cookie)(cookie)(cookie)"`.

`deposit(n)` should add n cookies to the cookie jar. If adding that many would exceed the cookie jar's capacity, `deposit` should raise a `ValueError`.

`withdraw(n)` should remove n cookies from the cookie jar. If there aren't that many cookies in the cookie jar, `withdraw` should raise a `ValueError`.

`capacity` should be a property that returns the cookie jar's capacity.

`size` should be a property that returns the number of cookies actually in the cookie jar, initially 0.

Your code should handle all edge cases appropriately and raise `ValueError` exceptions when required.

2 Giant Squid

Solve the **Advent of the Code 2021** day 4 problem. Try to use OOP concepts that you learnt in the class.

<https://adventofcode.com/2021/day/4>